Model answers

Question 1

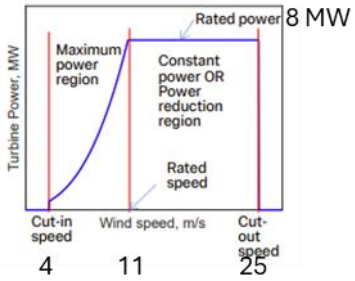
1.a.i)

$$\eta_{TOT} = \frac{P_e}{P_a} = \frac{P_e}{\frac{1}{2}\rho \cdot \pi \cdot R^2 \cdot v_w^3} = \frac{8 \cdot 10^6}{\frac{1}{2} \cdot 1.22 \cdot \pi \cdot 85^2 \cdot (11)^3} = 0.43$$ [2]

1.a.ii)

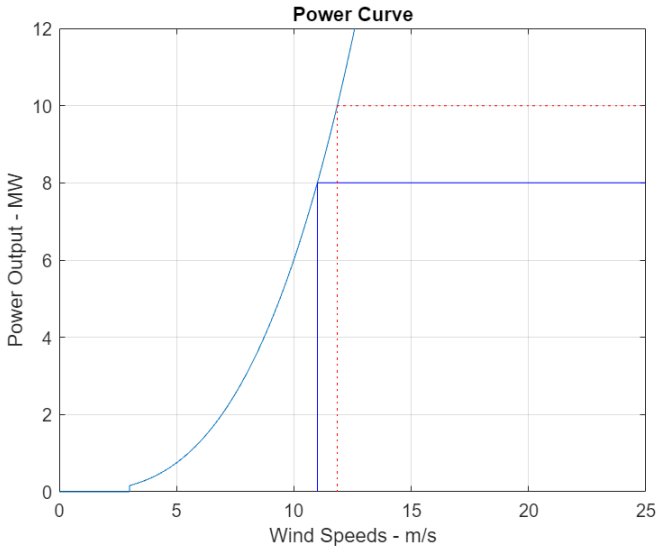
The efficiency calculated is lower than the Betz limit (16/27). This is expected because the total efficiency is the product of three efficiencies: aerodynamic efficiency, gearbox efficiency, and generator efficiency. In lectures, we estimated a range of total efficiency between 0.24 and 0.46. Therefore, while the total efficiency calculated in i) is lower than the Betz limit, it is high when the system is considered. [2]

1.a.iii)



[2]

1.a.iv)



1 mark for graph without new rated wind speed

[2]

1.a.v)

Rated speed: $v_w^3 = \frac{P_e}{\frac{1}{2}\rho \cdot \pi \cdot R^2 \cdot \eta_{TOT}} = 11.88 \frac{m}{s}$ [2 marks for rated speed; 1 from 1.a.iv) and 1 from this question]

The wind turbine will generate electricity at the rated power when the wind speed is between $11.88\frac{m}{s}$ and $25\frac{m}{s}$. Hence, the probability that the wind speed will be between these two speeds is

$$e^{-\left(\frac{11.88}{8}\right)^{2.2}} - e^{-\left(\frac{25}{8}\right)^{2.2}} = 0.0919277$$

Given that there are ~8,760 hours in one year, the wind turbine will generate at the rated power for ~806 hours in a year. [2]

1.b.i)

$$GHI = DNI \cdot \cos(\vartheta_z) + DHI$$
$$\vartheta_z = \varphi - \delta = 51 - (-23.5) = 74.5°$$
$$DNI = \frac{GHI - DHI}{\cos(\vartheta_z)} = \frac{0.171 - 0.113}{\cos(74.5\ ^\circ)} = 0.217\ kW/m^2$$

The two sources appear inconsistent. [2]

1.b.ii)

$$capacity\ factor = \frac{1{,}438\ kWh}{1\ kW \cdot 8760\ h} = 0.16$$ [2]

1.b.iii)

The double axis tracking system will be receiving 1,752kWh/m2 because the double-axis tracking system positions the panel perpendicular to the sun rays. Hence,

$$A = \frac{1438\ kWh}{1{,}752\ kWh/m^2 \cdot 0.25} = 3.28\ m^2$$ [2]

1.b.iv)

The single-axis tracking system cannot generate more than the double-axis tracking system. Hence, the capacity factor of the single-tracking system cannot be higher than 16%. [2]

1.b.v)

The single axis tracking system will generate at least the same amount of electricity as a horizontal system. The horizontal system will generate
$1{,}203 \cdot 0.25 \cdot 3.28 = 986kWh$.
Hence, the lower bound for the capacity factor will be $\frac{986\ kWh}{1\ kW \cdot 8760\ h} = 0.11$ [2]

Question 2

2.a)

Each CCGT can provide reserve 250 MW (= capacity (600MW) – minimum stable generation (350MW)). In order to deliver 6,500MW of reserve, 26 CCGT units are needed: 26x250 = 6,500 MW. These 26 CCGT units will also generate power: 26x300 = 7,800 MW (while delivering 6.5GW of reserve). [2]

Total power output:

- Must run Nuclear 8,400 MW
- CCGT: 7,800 MW
- Wind: 12,000 MW
- TOTAL: 28,200 MW

Therefore, wind power of 3,200 MW will need to be curtained. [1 mark]

This will increase the generation costs, as CCGT are running (burning gas), while wind, at zero marginal costs is curtailed.

This will also increase emissions as wind is curtailed and gas plant is operating ; also operation of part loaded CCGT will increase emissions due to reduction in efficiency.

[1 mark]

2.b Where there are no local measurements available, the network conditions are assessed using the controller's State Estimation module.

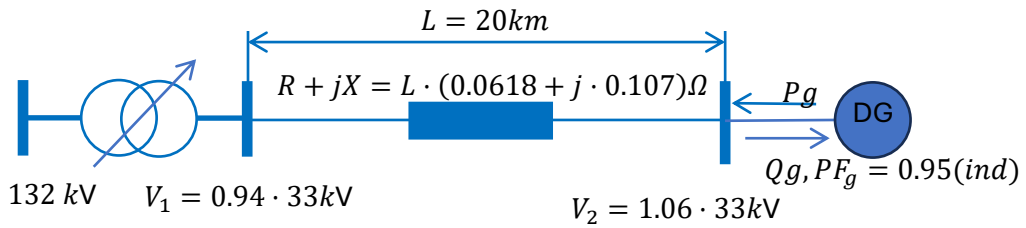
[Description along above or similar for 0.5 marks]

Pseudo measurements are used as input to state estimation to improve the estimation accuracy. Those are not the actual measurements but are used as if they were. It could be, for example, sum of all currents entering a node/bus is zero.

[Description along above or similar for 0.5 marks]

2.c.i)

Sketch of the considered system



[1 mark]

2.c.ii)

Derivation of the equation for the maximum active power of a distributed generation limited by the voltage rise effect

$$V_2 = V_1 + \frac{(R \cdot P_g - X \cdot Q_g)}{V_2}$$

$$Q_g = \sqrt{S_g^2 - P_g^2} = \sqrt{\frac{P_g^2}{PF_g^2} - P_g^2} = P_g \sqrt{\frac{1}{PF_g^2} - 1}$$

$$P_g^{max} = \frac{\left(V_2^{max} - V_1^{min}\right) \cdot V_2^{max}}{R - X\sqrt{\frac{1}{PF_g^2} - 1}}$$

[2 marks]

$$R + jX = 20 \cdot (0.0618 + j \cdot 0.107) = (1.236 + j \cdot 2.14)\Omega$$

$$P_g^{max} = \frac{(1.06 \cdot 33 - 0.94 \cdot 33) \cdot 1.06 \cdot 33}{1.236 - 2.14\sqrt{\frac{1}{0.95^2} - 1}} = 260.08MW$$

[1 mark]

2. d. i)

Capacity outage probability table for the generation plant is

Capacity in service, MW	**Probability**
10	0.9 x 0.8 x 0.8 = 0.576
7.5	2 x 0.9 x 0.8 x (1-0.8) = 0.288
5	0.9 x $(1-0.8)^2$ + (1-0.9) x 0.8^2 = 0.1
2.5	2 x (1-0.9) x 0.8 x (1-0.8) = 0.032
0	(1-0.9) x $(1-0.8)^2$ = 0.004

The total probability for all eight possible capacity states is 1.

[4 marks; subtract 1 mark for any missing capacity in service and wrong probability up to a maximum of subtracted 4 marks]

2.d.ii)

Energy not supplied (ENS) for a system state i, ENS_i, is calculated from the profile in the following way

$$ENS_i = \begin{cases} (10 - C_i) \cdot 240 & 7.5 \leq C_i \leq 10 \\ (10 - 7.5) \cdot 240 + (7.5 - C_i) \cdot 1{,}200 & 5 \leq C_i \leq 7.5 \\ (10 - 7.5) \cdot 240 + (7.5 - 5) \cdot 1{,}200 + (5 - C_i) \cdot 2{,}880 & 0 \leq C_i \leq 5 \end{cases}$$

Where C_i is system state capacity in. Profile from the figure 2.2 is scaled such that peak is equal to the total generation capacity i.e. 10 MW.

Expected energy not supplied (EENS) is equal to $ENS \cdot Probability$

Capacity, MW	**Probability**	**ENS, MWh**	**EENS, MWh**
10	57.6%	0	0
7.5	28.8%	600	172.8

5	10%	3,600	360
2.5	3.2%	10,800	345.6
0	0.4%	18,000	72
Total	**100.0%**		**950.4**

The total EENS is 950.4 MWh.

[5 marks; subtract 2 marks if it is not clearly explained how ENS and EENS are calculated; subtract 1 mark for any wrong pair ENS/EENS, including total EENS up to a maximum of subtracted 3 marks]

2.d.iii)

Comparing the total EENS and ENS state values in the above COPT, the firm capacity is between 5 and 7.5 MW.

[1 mark for finding a range where firm capacity could be found either from the COPT as mentioned above or from the appropriate load duration curve]

$$EENS = 950.4 = (10 - 7.5) \cdot 240 + (7.5 - C) \cdot 1{,}200$$

$$C = 7.5 - \frac{950.4 - (10 - 7.5) \cdot 240}{1{,}200} = 7.208MW$$

$$\frac{C}{10} = 0.7208 = 72.08\%$$

F Factor, of generating units, is 72.08%. Firm capacity is 7.208 MW.

[1 mark; calculation of value of firm capacity in MW is sufficient for the mark]

Question 3

3.a)

Building internal set temperature, K
Building external temperature, K
Other weather factors, e.g. solar gain, wind, K
Fabric (walls, windows, etc) thermal transmittance, W_{th}/m^2K
Air changes or permeability, $m_3/m_2.h$ @ 50Pa specified pressure
Other heat sources, e.g. occupants, appliances, etc [2]

3.b.i) ASHP transfers thermal energy from the outside air to the area to be heated. There are 4 stages to the cycle, and these comprise:
Evaporation - The refrigerant passes through a heat exchanger where it evaporates absorbing heat in the process.
Compression - The heated refrigerant is then compressed to higher pressure using an electrically driven compressor.
Condensation - The compressed refrigerant passes through another heat exchanger located in the space to be heated and condenses, changing state back to a liquid and rejecting heat in the process.
Expansion – The liquid refrigerant passes through an expansion device lowering its pressure prior to returning to the evaporator. [4]

3.b.ii) An ASHP can have an efficiency of greater than 100% as it excludes the energy absorbed from the air, i.e. it only includes the electrical energy for the heat pump's electrical demand. [1]

3.b.iii) A hybrid ASHP comprises a ASHP with an additional heat source such as a condensing gas boiler. This permits a smaller, lower cost heat pump to be used which may be a better match for the normal space heat demand of the building as it can operate at a higher efficiency. The condensing gas boiler operates when demand is high due to cold weather, for example, but it can also supply hot water demand if needed. [1]

3.b.iv)

Space heat demand (W_{th})
= ($T_{int} - T_{ext}$) * Average thermal transmittance * Floor area
= (20-0) *3.75*100 = **7500W_{th} or 7.5kW_{th}**

Where:
T_{int} = Building internal set temperature, °C
T_{ext}= Building external set temperature, °C

The householder should choose the **7.5kW_{th} heat pump**.

Assumptions:
Estimate of heat demand does not take account of:
other weather factors such as wind, solar gain.
building air changes
other heat sources [4]

3.b.v)

Space heat demand (W_{th}) = ($T_{int} - T_{ext}$) * Revised average thermal transmittance * Floor area

Rearranging:

Revised average thermal transmittance = Space heat demand (W_{th}) / (($T_{int} - T_{ext}$) * Floor area)

= 7500Wth/((20-(-5))*100) = 3.0W_{th}/m^2K

Improvement in building efficiency, % = (1-3.0/3.75)*100 = **20%** **[2]**

3.b.vi)

Typical annual building space heat demand (MWh_{th})

= ($T_{int} - T_{ext}$) * Average thermal transmittance * Building floor area*Nos of days heating is on in a year*Nos of hours heating is on during the day

= (20-7) *3.0*100*200*10 = **7,800,000Wh_{th}** or **7.8MWh_{th}** **[4]**

3.c.i)

The purpose of the CCC is to:

- advise UK Government and Devolved Administrations on emissions targets and report to Parliament on progress made in reducing greenhouse gas emissions and preparing for climate change
- provide independent advice on setting and meeting carbon budgets and preparing for climate change
- monitor progress in reducing emissions and achieving carbon budgets and targets
- conduct independent analysis into climate change science, economics and policy
- engage with a wide range of organisations and individuals to share evidence and analysis [1]

3.c.ii)

The UK Government amended the target from 80% reduction of greenhouse gases by 2050 to 100% reduction to achieve Net Zero GHG emissions. [1]

Question 4

4.a) Based on typical vehicle usage patterns and specific features of the EV technology, EV loads are well suited to provide flexibility in order to support system operation, for the following reasons:

- The additional energy requirements for EV charging are relatively modest compared to the current electricity demand. Even if the entire UK light vehicle fleet switches to electricity, the total demand (according to the assumptions made above) would not exceed 53 TWh annually, which is broadly equivalent to 15% of today's annual electricity system demand.
- Driving times generally associated with the majority of small passenger vehicles are relatively short. The proportion of non-stationary vehicles rarely exceeds 20% during a typical day.
- Given that EV batteries have relatively high power ratings and significant amount of inherent storage when considered in aggregation, they could potentially be used as flexible demand resource to enable more efficient system operation.

There is considerable flexibility regarding the time when the vehicles can be charged and this can provide significant benefits both to the operation and design of distribution and transmission networks as well as to the efficient dispatch and utilisation of generation.

Uncontrolled EV charging might have a significant impact on peak system demand, as the total demand during peak periods may rise significantly for high penetrations of EV due to synchronisation of peak EV load and the original system peak demand. This would require significant additional capacity of the system infrastructure (generation and networks) in order to maintain the system security at an acceptable level. The increase in capacity requirements driven by higher peak demand will be disproportionately higher than the increase in energy requirements for EV charging.

In contrast, alternative (smart) EV charging strategies can result in very modest peak increase compared to the system without EV demand, while still delivering the same energy to EVs that is required to complete the same journeys.

(1 mark for the explanation of EV flexibility)

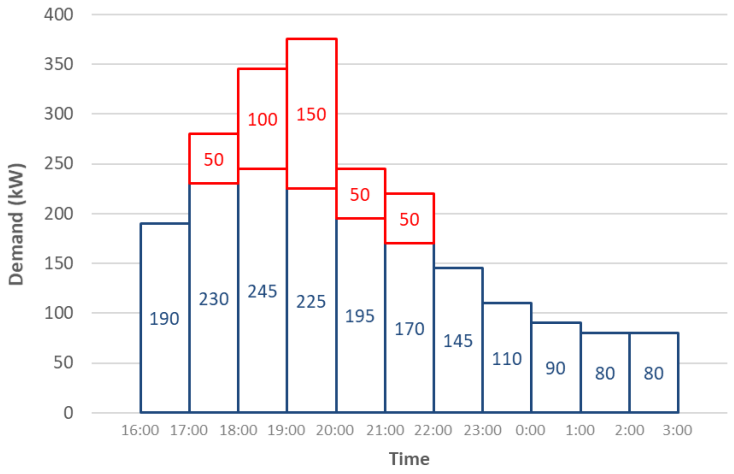
(1 mark for the explanation of the impact of EVs on energy and peak demand)

4.b) EVs in groups 1, 2 and 3 require 5, 10 and 15 kWh per vehicle, respectively, to charge their batteries to 100%. Respective charging times are 1, 2 and 3 hours when charging at rated power of the charger (5 kW). Multiplying the charger rating with the number of EVs gives the charging power for all EVs in group 1, 2 and 3 of 50 kW, 100 kW and 50 kW, respectively.

i) If all EV users plug in their vehicles to charge immediately upon arrival, and charge them at maximum capacity until their batteries are full, the resulting demand for each EV group as well as total demand are given in the table below.

Hour	Baseline	EV #1	EV #2	EV #3	Total EVs	Total demand
16:00	190					190
17:00	230	50			50	280
18:00	245		100		100	345
19:00	225		100	50	150	375
20:00	195			50	50	245
21:00	170			50	50	220
22:00	145					145
23:00	110					110
00:00	90					90
01:00	80					80
02:00	80					80

The new demand diagram including EVs is shown here:

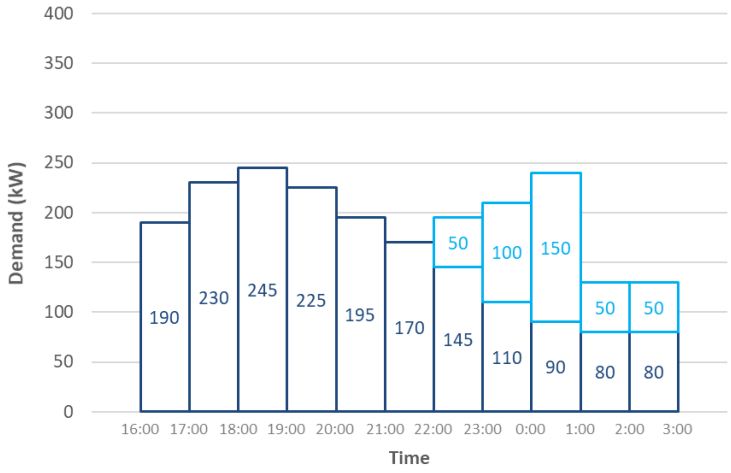


(1 marks for the correct diagram)

Peak demand is achieved in the hour between 19:00 and 20:00, at the level of **375 kW**.

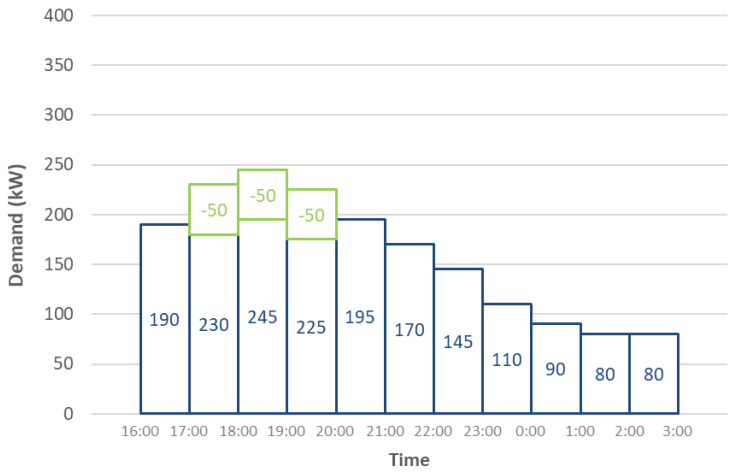
(1 marks for the correct peak demand value)

ii) To avoid increasing total peak demand above the baseline value, all EVs need to delay their charging by at least **5 hours**. The resulting demand diagram after shifting EV charging by 5 hours is shown below, confirming that the peak demand is maintained at 245 kW.



(3 marks)

iii) Discharging EVs in group #1 can inject 10*5 = 50 kW of power back into the grid. Doing this for 3 hours between 17:00 and 20:00 would require 15 kWh per EV, which is less than the assumed SoC on arrival (20 kWh), implying that this service is feasible. V2G reduces the demand in all three hours between 17:00 and 20:00 by 50 kW, so that the new peak demand occurs between 18:00 and 19:00 (as well as between 20:00 and 21:00) and is equal to 195 kW. Given that all EVs can be charged after 3:00am without any impact on the total peak demand, the new net peak demand is equal to **195 kW**. The resulting demand diagram is shown below.



(3 marks)

4.c.i)

Sunamp: Thermal storage, Latent heat
StrataStore: Mechanical storage, Potential Energy [4]
1 mark for each correct category

4.c.ii) [2]

StrataStore Advantages
- Low specific energy capacity cost
- Independent sizing of energy capacity and power capacity
- Long lifetime

StrataStore Disadvantages
- Cost-efficient underground CAES plants are geographically limited by the availability of caverns

- Low energy density (~4 kWh/m3)
- Only economic at large scale (> multiple hundred MW) and long discharge duration (> 4 hours)

0.5 mark per correct advantage, disadvantage.

4.c.iii)

The storage system is ‘charging’ during compression and discharging during expansion.
The round-trip efficiency is $0.8 \cdot 0.8 = 0.64$
This round-trip efficiency is relatively small compared to batteries where the round-trip efficiency is ~85%. [2]

4.c.iv)

I expect that the batteries will cycle more often because they have higher round-trip efficiency. However, given the cycle-related degradation for batteries, I expect that cycling behaviour will be different with respect to the depth of discharge, with batteries having more shallow cycles than the StrataStore Project. [2]